Project Initiation Document

Smart Ambulance Green Corridor System

1. Project Overview

Item	Details
Title	Smart Ambulance Green Corridor System (single 4-way junction)
Objective	Detect an approaching ambulance and give it a green light using only digital logic ICs (counters, priority encoder, FSM).
Scope	One junction (N/E/S/W), 12 signal LEDs, toy-scale road model. Multi-junction corridors are out of scope.
Constraints	No Arduino or microcontroller; 74HC-series/CMOS ICs and 555 timers only.
Approach	Push-button request switch per direction (simulating ambulance arrival), latched and debounced; priority encoder picks the winner; FSM controls lights; counter handles timing.

2. Deliverables

- Working hardware model on a road-layout board
- State diagram and state table
- Circuit schematic (block level and pin level)
- Test report with results for each test case
- Project report and demonstration

3. Success Criteria

- Normal cycle runs with correct timing (green 8 s, yellow 2 s, all-red 1 s)
- Ambulance direction gets green within 3 s of detection (after current yellow, if mid-cycle)
- Junction returns to normal after clearance or after the 10 s timeout
- Simultaneous requests resolved by fixed priority N > E > S > W
- No two conflicting greens ever lit at the same time

4. Risks and Mitigation

Risk	Mitigation
Switch contact bounce	RC filter + 74HC14 Schmitt trigger, then NAND SR latch on each request line
Operator releases button early	SR latch holds the request until the clear switch is pressed
Clock glitches / metastability	D flip-flop synchronizers on the system clock
Power supply noise	7805 regulator, 0.1 uF decoupling on every IC
Clear switch not pressed	Timeout counter forces return to normal after 10 s
Wiring errors on breadboard	Build and test block by block; simulate first

5. Project Plan

Phase	Activity
1	Finalize requirements, state diagram, and equations
2	Simulate in Logisim / Proteus
3	Breadboard individual blocks (clock, counter, FSM, switch inputs)
4	Integrate and test full system
5	Solder on perfboard, mount on road model
6	Final testing, report, demonstration